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Prato reaction derived polythiophene/C₆₀ donor–acceptor double cable polymer, fabrication of photodetectors and evaluation of photocurrent generation†Sunniya Iftikhar,^a Saba Aslam,^a Nauman Zafar Butt,^b Raja Shahid Ashraf^c and Basit Yameen  [✉]

Herein, a side chain engineered polythiophene copolymer based organic semiconductor polymer (OSP) with pendant aldehyde groups was synthesized by a combination of Grignard Metathesis (GRIM) polymerization and post-synthetic modifications. The pendant aldehyde groups of OSP (electron donor-D) were covalently conjugated to the pristine C₆₀ (electron acceptor-A) via 1,3-dipolar cycloaddition reaction (i.e., Prato reaction) between the pristine C₆₀ and azomethine ylide formed *in situ* as a result of the reaction of pendant aldehyde groups and sarcosine. The OSP with pendant aldehyde groups and its covalent conjugate with pristine C₆₀ were fully characterized using FTIR and ¹H-NMR spectroscopies. Fluorescence spectroscopic analysis of the synthesized D–A double cable covalent hybrid and physical hybrid derived from the physical mixing of the synthesized OSP and C₆₀ revealed a higher fluorescence quenching in the case of the covalent hybrid. This novel precisely designed D–A hybrid material was employed as the photoactive layer in an interdigitated gold electrode based photodetector where a significant enhancement in photocurrent (>2.5 times) was observed in the devices consisting of photoactive layer derived from the D–A covalent hybrid. This work offers a step forward in the systematic development of D–A covalent hybrids with a precise molecular design to fabricate efficient photovoltaic devices.

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rsc.li/materials-c**Introduction**

Organic semiconducting polymers (OSPs) are being widely explored for application in diverse optoelectronic technologies such as photovoltaic devices, light-emitting diodes, and field-effect transistors.^{1–4} Compared to their inorganic counterparts, OSPs offer ease of solution processability^{5–9} as well as higher transparency and flexibility of the resulting thin films, and lighter weight stemming from their comparatively lower bulk density that ultimately leads to a lower device fabrication cost.^{10–17} In addition, OSPs offer a unique opportunity of precise modulation of their physical, chemical and electronic properties by carefully controlling their molecular architecture during synthesis or as a post-synthetic modification endeavour.

This has led to the extensive optimization efforts producing a wide variety of OSPs with control over their structure–property relations aligned with the requirements of a desired optoelectronic application.^{18–20} The organic photovoltaic devices work on the photovoltaic effect's basic mechanism involving the photo-generation of excitons (electron–hole pairs) that are separated to generate photocurrent.²¹ Due to their large binding energy (0.3–0.1 eV) and limited diffusion length (~10 nm), the efficient generation of excitons, their diffusion, separation into holes and electrons followed by charge transport and collection constitute the main challenges in enhancing the efficiency of photocurrent generation in OSP based photovoltaic devices and their subsequent real life applications.^{22,23} To overcome these challenges, photovoltaic devices comprising a photoactive layer derived from the physical hybrids of π -conjugated OSPs as electron donors (D) and fullerenes (e.g., C₆₀) or non-fullerene high electron affinity acceptors (A) have been extensively investigated under bulk heterojunction (BHJ) arrangement.²⁴ Extensive research efforts are being employed to synthesize novel OSP electron donors with the desired optical (e.g., band gap) and hole-transport properties. In this context, thiophene based semiconductor polymers are the most widely explored electron donors. Attempts have been made to

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tune the band gap of OSPs by installing a wide variety of band gap modulating functionalities^{25–30} within the main π -conjugated backbone of OSPs. In addition, substituting electron-withdrawing groups and alkyl chains on the π -conjugated OSPs has been demonstrated to influence their energy levels, enhance overall crystallinity, increase solubility, improve self-assembly of molecules, and enhance photocurrent generation.^{26,31–33} Besides modification of the electron donors, structural modifications of electron acceptor moieties have also been widely investigated. A number of fullerene derivatives have been synthesized primarily to improve their solubility in common solvents, modulate their electronic structure, and improve compatibility with the electron donors to enhance the photocurrent generation. In this context, a variety of chemical reactions have been employed to functionalize fullerenes.³⁴ Cycloaddition reactions (*e.g.*, [1+2] cycloaddition reactions with carbenes and nitrenes, [3+2] cycloaddition reactions with azomethine ylide known as the Prato reaction, Diels–Alder cycloaddition reactions), reaction with radicals, the addition of nucleophiles (*e.g.*, Bingel reaction), and reactions with electrophiles have been extensively reported.^{35–40}

Besides the optimum physicochemical and electronic properties of D and A materials, an optimum self-assembly of their physical hybrids is the key to achieving an improved D–A electronic interaction.^{34,41,42} The self-assembly and D–A electronic interaction are influenced by the choice of solvent used during film fabrication and solvent assisted annealing processes as well as the choice of temperature, thermal annealing of the fabricated film.^{43–45} These device fabrication parameters assist in the optimization of the D–A interaction while at the same time they also underscore the challenge of the instability of BHJ derived from the physical hybrids towards environmental stresses.⁴⁶ The environmental stresses can trigger separation of D and A molecules followed by the formation of large separate domains of D and A materials, which drastically reduces the device performance. Consequently, several strategies are being explored to stabilize the BHJ. Besides their physical incorporation in the BHJs, fullerenes and their derivatives have also been covalently conjugated to OSPs in several attempts to control the D–A electronic interaction and enhance the generation of photocurrent. In this context, fullerenes have been either directly conjugated to the backbone of the OSPs (homopolymer) or conjugated to the polyolefinic block of OSP-co-polyolefinic block copolymer or conjugated as end groups to the π -conjugated organic semiconductor oligomers.^{47–54} Based on the reported literature and our own previous work, we believe that the OSPs capable of covalently reacting with fullerenes can provide unprecedented control over electronic interaction between donor OSPs and fullerenes. This donor–acceptor molecular intimacy ultimately opens a new avenue to regulating and improving the performance of devices derived from the OSP based hybrid materials. In our ongoing quest for novel OSP–fullerene covalent conjugates with controlled molecular architecture and enhanced photocurrent generation, herein, we report a novel side chain engineered polythiophene capable of covalently conjugating to pristine C₆₀ *via* Prato reaction and produce a sort of double cable polymer. The side chain engineered fully conjugated polythiophene with pendant aldehyde groups was synthesized

via a combination of Grignard Metathesis (GRIM) polymerization and controlled post-synthesis side chain modifications. The synthesized polythiophene/C₆₀ covalent donor–acceptor conjugate was characterized for its physicochemical properties and was employed to fabricate the organic photodetector (OPD) to evaluate the photocurrent generation. It is worthwhile highlighting here that the photoactive layer comprising of D–A BHJ constitutes the heart of photovoltaic devices and the so far elaborated fundamental principles are applicable to all the devices relying on the photovoltaic effect. OPDs and organic solar cells (OSCs) are two prime examples of such devices. In the current work, we have resorted to OPDs to test our hypothesis and demonstrate the effectiveness of our approach. The photocurrent generation capability of the novel polythiophene/C₆₀ covalent conjugate was superior to the photocurrent generation capability of the physical hybrid of poly-3-hexylthiophene (P3HT) and C₆₀. This work is a systematic progression of the endeavours focused on the synthesis of covalent D–A conjugates with controlled electronic communication and a step towards achieving efficient BHJs.

Experimental section

Materials

Anhydrous tetrahydrofuran (THF) ($\geq 99.9\%$), anhydrous dichloromethane (DCM) ($\geq 99.8\%$), 3-thiophenemethanol (98%) (**1**), *N*-bromosuccinimide (NBS) (99%), *p*-toluene sulfonic acid ($\geq 98.5\%$), dichloromethane (DCM) (99.0%), 3,4-dihydro-2*H*-pyran (97%), tertiary butylmagnesium chloride (2 M in diethyl ether), [1,3-bis(diphenylphosphino)propane]dichloronickel(III) [Ni(dppp)Cl₂] (97%), triethylamine (99.5%), sodium bicarbonate (NaHCO₃) (99.7%), sarcosine (*N*-methyl glycine) (98%), fullerene-C₆₀ (99.5%), and sodium borohydride (NaBH₄) ($\geq 96\%$) were purchased from Sigma Aldrich. 2,5-Dibromo-3-hexylthiophene ($\geq 97.0\%$) was purchased from Tokyo Chemical Industry (TCI, Japan). (2,5-Dibromothiophen-3-yl)methanol (**2**) and 2-((2,5-dibromothiophen-3-yl)methoxy)tetrahydro-2*H*-pyran (**3**) were synthesized according to literature procedures.⁵⁵ 4-Formylbenzoyl chloride was synthesized from 4-formylbenzoic acid.⁵⁶ Hexane, ethyl acetate, methanol, and chloroform were used after distillation. Aluminum sheets (Merck) coated with silica gel 60 F254 were used for thin layer chromatography (TLC).

Instrumentation

Gas chromatography-mass spectrometry (GC-MS) was performed on a Trace 1300 Gas Chromatograph coupled with an ISQ mass spectrometer. Attenuated Total Reflection Fourier Transform Infrared (ATR-FTIR) spectroscopy was performed on an ALPHA FTIR spectrometer from Bruker. Nuclear magnetic resonance (NMR) spectroscopy was performed on a Bruker ASENTM 600 NMR spectrometer. UV/Visible spectra were recorded by using a SCHIMADZU UV-1800 spectrophotometer whereas fluorescence emission spectra were recorded on an Agilent Technologies Carry Eclipse Fluorescence Spectrophotometer. Gel permeation chromatography (GPC) measurements were performed on an Agilent Technologies 1260 Infinity II GPC/SEC system. This instrument

consists of an auto-sampler, one PL gel 5 μm guard column (50 $\times$ 7.5 mm) followed by two PL gel 5 μm Mixed-D columns (300 $\times$ 7.5 mm); and viscometer and refractive index detectors. THF was used as the eluting solvent with a flow rate of 1 mL min⁻¹ at 35 °C. Moreover, the linear PS standards were used for the calibration of the GPC system having a range of 160 to 3.64 $\times$ 10⁵ g mol⁻¹.

For device fabrication, a Karl Suss MJB 3 Mask Aligner with a 350 mercury arc lamp having 365 nm and 405 nm wavelength UV filters was used to develop interdigitated patterns on the glass substrates and VTS, South Korea Magnetron's Sputtering machine from VTS Korea was used to coat titanium and gold layers on the device pattern containing glass substrates. Current-voltage profiles of photodetectors were recorded using a Keithley 2461 Source Meter.

GRIM polymerization of 2,5-dibromo-3-tetrahydropyranoxy-methylthiophene 3 and 2,5-dibromo-3-hexylthiophene to synthesize random copolymer 4

1.03 g (2.94 mmol) of 3 and 2.84 g (8.82 mmol) of 2,5-dibromo-3-hexylthiophene were dissolved in 116 mL of anhydrous THF under inert conditions. 5.89 mL (11.8 mmol) of 2 M ether solution of *tert*-butylmagnesium chloride was added into the monomer solution. The reaction mixture was refluxed under an inert atmosphere (argon gas) for 90 minutes. The reaction mixture was cooled to room temperature followed by the addition of 127 mg (0.23 mmol) of Ni(dppp)Cl₂. The polymerization was allowed to proceed at room temperature for 10 minutes and then poured into ice-cold methanol to precipitate out the polymer. The precipitated polymer was filtered and subjected to Soxhlet extraction (overnight) with methanol to remove the impurities. The random copolymer 4 (M_n = 12 300 Da, D = 1.1) was collected in chloroform during Soxhlet extraction. The solvent was removed *via* rotary evaporator and the obtained polymer was dried overnight under vacuum at room temperature.⁵² ¹H-NMR (CDCl₃, δ ppm): signals originated from repeat units incorporated in copolymer 4 through monomer 2,5-dibromo-3-hexylthiophene and monomer 3; signals originated from 2,5-dibromo-3-hexylthiophene derived repeat unit: 0.85 (3H), 1.25–1.7 (8H), 2.75 (2H), 6.94 (1H); signals originated from monomer 3 derived repeat unit: 1.25–1.5 (6H), 3.53 (1H), 3.9 (1H), 4.5–4.7 (2H), 4.78 (2H), 7.1 (1H).

Deprotection of pendant tetrahydropyranyl ether groups of random copolymer 4 to give random copolymer 5 with pendant hydroxyl groups

0.41 g of random copolymer 4 was dissolved in 110 mL of THF, followed by the slow addition of 12 mL of concentrated HCl. The reaction mixture was refluxed overnight. The reaction mixture was then cooled down to room temperature followed by the addition of 19 g of NaHCO₃ in small fractions to neutralize the HCl. The suspension obtained was filtered and the filtrate was concentrated on a rotary evaporator. The concentrated filtrate was poured into ice-cold methanol to precipitate out the polymer. The polymer precipitates were collected *via* filtration and subjected to Soxhlet extraction (overnight) with methanol.

The random copolymer with pendant hydroxyl groups was collected by replacing methanol with chloroform during Soxhlet extraction. Random copolymer 5 was obtained by removing the chloroform on the rotary evaporator and subjecting it to drying overnight under a vacuum.⁵² ¹H-NMR (CDCl₃, δ ppm): signals originated from the repeat units incorporated into copolymer 5 through monomer 2,5-dibromo-3-hexylthiophene: 0.85 (3H), 1.25–1.7 (8H), 2.75 (2H), 6.94 (1H); signals originated from the repeat units incorporated into copolymer 5 through monomer 3 after deprotection of the protected –OH groups: 4.5–4.7 (2H), 7.1 (1H).

Reaction of pendant hydroxyl groups of random copolymer 5 with 4-formylbenzoyl chloride to give random copolymer 6 with pendant benzaldehyde groups

250 mg of random copolymer 5 was dissolved in 20 mL of anhydrous DCM under an inert atmosphere followed by the addition of 167 μL (1.2 mmol) of triethylamine. The reaction mixture was placed in an ice bath. 404 mg (2.4 mmol) of 4-formylbenzoyl chloride was separately dissolved in 10 mL of anhydrous DCM and the resulting solution was dropwise added to the ice cooled reaction mixture under an inert atmosphere. The reaction mixture was stirred for 30 minutes after which the ice bath was removed and the reaction mixture was stirred overnight at room temperature. The reaction was subjected to washing with 0.5 M HCl (three times), 0.5 M NaHCO₃ (three times) and finally brine solution (three times). The organic layer was dried over anhydrous MgSO₄, and filtered to collect the filtrate. The solvent of the filtrate was removed on a rotary evaporator to give polymer 6 that was dried overnight under vacuum at room temperature. ¹H-NMR (CDCl₃, δ ppm): signals originated from the repeat units incorporated into copolymer 6 through monomer 2,5-dibromo-3-hexylthiophene: 0.85 (3H), 1.25–1.7 (8H), 2.75 (2H), 6.94 (1H); signals originated from the repeat units incorporated into copolymer 6 through monomer 3 bearing pendant benzaldehyde: 4.52 (1H), 4.72 (1H), 7.1 (1H), 7.52–8.3 (4H), 10.06 (1H).

Prato reaction of random copolymer 6 with fullerene to give covalent hybrid 7

40 mg of polymer 6 was dissolved in 10 mL of toluene under an inert atmosphere. 4.5 mg (0.05 mmol) of sarcosine (*N*-methyl glycine) was added to the solution. 28 mg of fullerene was separately dissolved in 5 mL of toluene and was slowly added to the reaction mixture. The reaction mixture was refluxed for 4 h at 120 °C under an inert atmosphere. After cooling down to room temperature, the reaction mixture was extracted with DCM. Subsequently, DCM extract was washed with 10% aq. HCl solution, then with a 1 M sodium bicarbonate solution and finally with brine solution. The organic layer was dried over anhydrous MgSO₄. The solvent was evaporated on a rotary evaporator and the resulting covalent hybrid 7 was dried overnight under vacuum at room temperature.^{57,58} ¹H-NMR (CDCl₃, δ ppm): signals originated from the repeat units incorporated into copolymer 7 through monomer 2,5-dibromo-3-hexylthiophene: 0.85 (3H), 1.25–1.7 (8H), 2.75 (2H), 6.94 (1H); signals originated from the

repeat units incorporated into copolymer 7 through monomer 3 after covalent conjugation with C₆₀: 2.26 (3H), 4.52 (1H), 4.72 (1H), 7.1 (1H), 7.52–8.3 (4H).

Preparation of physical hybrid 8 of random copolymer 6 and fullerene

40 mg of copolymer 6 was dissolved in 10 mL of toluene under inert conditions. 28 mg of fullerene was separately dissolved in 10 mL of toluene and added dropwise to the solution of polymer 6. The reaction mixture was kept on stirring at room temperature for 4 h. The solvent was then removed on a rotary evaporator to give physical hybrid 8.

UV/Vis and fluorescence spectroscopies

For recoding UV/Vis absorption and fluorescence spectra, 2 mg of the given samples (donor polymer, donor–acceptor covalent hybrid, donor–acceptor physical hybrid and pure fullerene) were dissolved in 4 mL of CHCl₃ at room temperature. UV/Vis absorption and fluorescence spectra were recorded on these solutions. The solutions were then diluted 100 fold and UV/Vis absorption and fluorescence spectra were recorded on the diluted solutions as well.

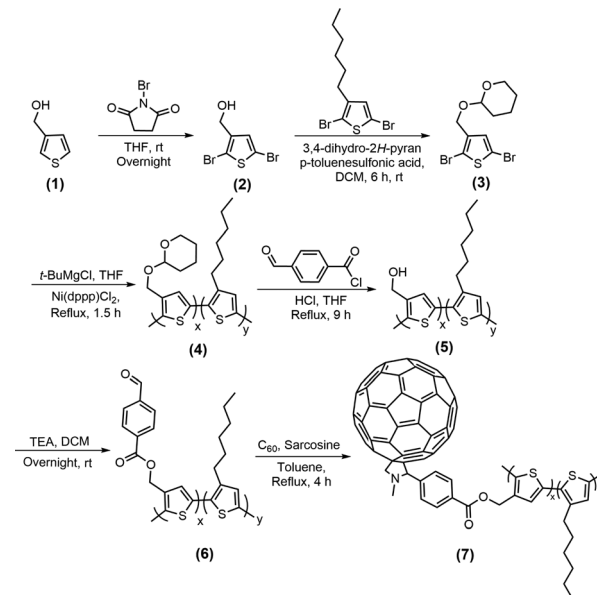
Device fabrication

In device fabrication, photolithography and lift-off methods were used to prepare interdigitated Au electrodes. During this process, a titanium layer of thickness 10 nm and Au layer of thickness 70 nm were deposited on glass substrates *via* a sputtering process. Each Au finger's width and the spacing between neighboring fingers were controlled to be 50 μm. The interdigitated electrodes were cleaned ultrasonically in acetone, water, isopropanol, and water sequentially. 2 wt% solutions of polymer 6, covalent hybrid 7 and physical hybrid 8 were prepared in chloroform. These prepared solutions were spin-coated at 2000 rpm for 30 s onto the interdigitated Au electrodes and then heated at 100 °C for 5 minutes.⁵⁹ The performance of the photodetector was measured by using a Keithley 2461 Source Meter. The flashlight was used to observe the effect of light on the active area. All the measurements were performed in air at room temperature.

Results and discussion

Polythiophene with pendant aldehyde groups was synthesized by a combination of GRIM polymerization and precisely controlled post-synthetic modifications. GRIM polymerization has emerged as a facile route to polythiophenes with controlled molecular weights, narrow polydispersities, and well-defined end as well as pendant groups. Employing GRIM polymerization, a statistical random copolymer 4 (GPC: *M*_n 12 300 Da, *D*: 1.1) bearing pendant tetrahydropyranyl ether groups was synthesized by employing 2,5-dibromothiophene (75 mol%) and 2,5-dibromothiophene compound (3, 25 mol%) as monomers (Scheme 1).⁵²

Acid-catalyzed deprotection of the pendant tetrahydropyranyl ether groups in 4 and subsequent esterification reaction of the



Scheme 1 Schematic illustration of the synthesis of the side chain engineered polythiophene based random copolymer by GRIM polymerization and post-synthetic modifications leading to the covalent hybrid 7 via the Prato reaction.

resulting pendant hydroxyl groups with 4-formylbenzoyl chloride resulted in polymer 6 with pendant aldehyde groups. The polythiophene copolymer 6 was covalently conjugated to fullerenes *via* a 1,3-dipolar cycloaddition reaction (Prato reaction) between fullerene and azomethine ylide formed *in situ* as a result of the reaction of aldehyde pendant groups of polythiophene copolymer and sarcosine (*i.e.*, *N*-methyl glycine).^{57,58}

The FTIR spectra of copolymer 4 with tetrahydropyranyl (THP) ether pendant groups and copolymer 5 with pendant hydroxyl groups obtained after the deprotection of the pendant THP ether groups of copolymer 4 are shown in Fig. 1a and b. The FTIR spectra of copolymer 6 with pendant benzaldehyde

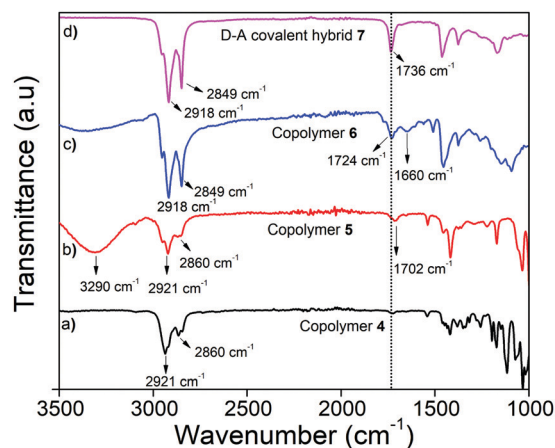


Fig. 1 FTIR spectra of (a) copolymer 4 with pendant THP ether groups, (b) copolymer 5 with pendant hydroxyl groups, (c) copolymer 6 with pendant aldehyde groups and (d) D–A covalent hybrid 7 obtained after Prato reaction between 6 and C₆₀.

groups and D–A covalent hybrid **7** obtained *via* Prato reaction of copolymer **6** and fullerene are depicted in Fig. 1c and d. In copolymer **4**, the hydroxyl groups were protected with THP ether groups; therefore, no prominent signal for hydroxyl groups was observed in the FTIR spectra of copolymer **4**. The pendant hydroxyl groups revealed in copolymer **5** as a result of deprotection of THP ether groups resulted in a conspicuous absorption at 3290 cm^{-1} due to the --OH stretching vibration (Fig. 1b). In the FTIR spectra of both the copolymers **4** and **5**, an absorbance signal at 1702 cm^{-1} was observed, which may be attributed to the weak aromatic overtones.^{60,61} The reaction of pendant hydroxyl groups of copolymer **5** with 4-formylbenzoyl chloride resulted in the appearance of absorption signals at 1660 cm^{-1} and 1724 cm^{-1} corresponding to C=O stretching of aldehyde and ester groups of the copolymer **6**. The successful covalent conjugation of copolymer **6** with fullerene *via* Prato reaction resulted in the complete disappearance of absorbance for the aldehyde group in the FTIR spectrum of covalent hybrid **7**. In addition, the absorbance for ester linkage in covalent hybrid **7** was observed to shift to a slightly higher wavenumber (1736 cm^{-1}). Absorbance signals at 2849 cm^{-1} and 2918 cm^{-1} in the FTIR spectra of polymers **6** and **7** correspond to the C--H stretching vibrations. Moreover, the characteristic peaks of different functional groups in copolymers **4**, **5**, **6** and **7** are collated in Table 1.

^1H -NMR spectroscopy was employed to characterize all the transformations depicted in Scheme 1. A comparison of the

integrals of the resonance signals for 'g' and 'h' protons of two differently substituted thiophene rings incorporated in the polythiophene chain revealed an incorporation of 17 mol% of monomer **3** in the random copolymer **6**. The deprotection of pendant THP ether groups followed by the esterification reaction of the pendant hydroxyl groups with 4-formylbenzoyl chloride was confirmed by the appearance of resonance signals associated with the protons of the benzene ring (k, l) of the 4-formylbenzoyl group appearing around 7.5–8.5 ppm and aldehyde proton appearing at around 10 ppm in the ^1H -NMR spectrum of **6** (Fig. 2). ^1H -NMR analysis also gives important information about structural regioregularity of the substituted polythiophenes. In the polymer chains, the head to tail (HT) and head to head (HH) arrangement of the repeat units can be estimated from the characteristic peaks of the α -methylene group. It can be observed (Fig. 2a) that the chemical shift in the range of 2.75 to 2.8 ppm is attributed to the head to tail configuration whereas the head to head configuration peak lies in the range of 2.5 to 2.6 ppm. The percentage of HT and HH content in the copolymer **6** calculated from the relative ratio of the integral of these peaks was found to be 87% and 13%, respectively. The regioregularity of copolymer **4**, **5** and **7** was also found to lie in the same range (Fig. S1, ESI[†]).

The integration of the characteristic 'k' and 'l' signals of copolymer **6** observed in the range of 7.5–8.5 ppm (Fig. 2a) was used to estimate the number of pendant 4-formylbenzoyl groups substituted on the copolymer backbone, which were found to be around 10 mol%. The covalent conjugation through the Prato reaction of copolymer **6** with C_{60} *via* its pendant aldehyde groups was ascertained by the disappearance of the resonance signal of the aldehyde proton that appeared around 10 ppm in the ^1H -NMR spectrum of the copolymer **6**. This signal completely disappeared in the ^1H -NMR spectrum of the covalent hybrid **7** (Fig. 2b), confirming the transformation of a pendant aldehyde group to a pyrrolidinofullerene group as a result of the Prato reaction. In addition, ^{13}C -NMR spectroscopic analysis also corroborated the synthesis steps depicted in Scheme 1. ^{13}C -NMR spectra of the pendant hydroxyl group with 4-formylbenzoyl chloride were confirmed by the appearance of

Table 1 Characteristic absorbances corresponding to important functional groups of copolymers **4**, **5**, **6** and **7**

Sample ID	Wavenumber (cm^{-1})	Corresponding functional group
Copolymer 4	2860, 2921	C–H stretching
Copolymer 5	3290	O–H stretching
	2860, 2921	C–H stretching
	1702	Aromatic overtones
Copolymer 6	2849, 2918	C–H stretching
	1724	C=O stretching (ester)
	1660	C=O stretching (aldehyde)
Copolymer 7	2849, 2918	C–H stretching
	1736	C=O stretching (ester)

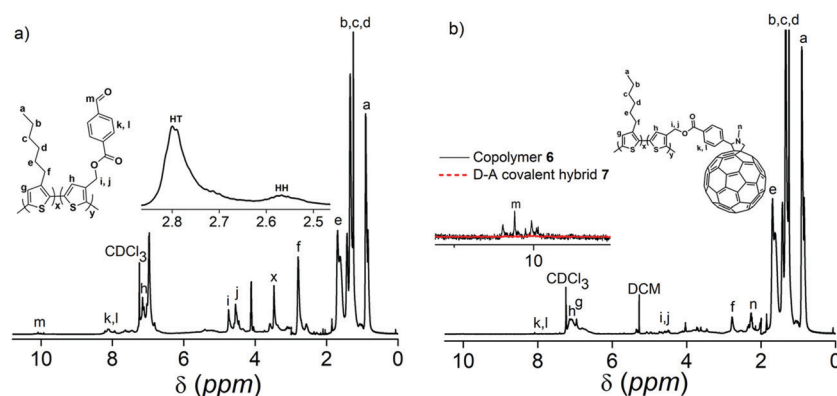


Fig. 2 (a) ^1H -NMR spectra of copolymer **6**, and (b) ^1H -NMR spectra of D–A covalent hybrid **7** along with the comparison of the regions of the spectra of **6** and **7** characteristic of an aldehyde proton.

characteristic signals associated with the carbonyl carbon of the ester group (r) appearing around 170 ppm and copolymers **4**, **5**, **6**, and **7** are provided in Fig. S2 in the ESI†. The esterification reaction of aldehyde carbon (s) appearing at around 205 ppm in the ^{13}C -NMR spectrum of **6** (Fig. S2c, ESI†). The signals (a'–f') in the range of 36–80 ppm validating the chemical transformation of the pendant aldehyde group of copolymer **6** into a pyrrolidinofullerene group (Fig. S2d, ESI†). The disappearance of the characteristic signal of the aldehyde carbon in copolymer **6** and the appearance of the characteristic signal for the C_{60} moiety of D–A covalent hybrid **7** at around 143 ppm (Fig. S2, ESI†) confirms that the Prato reaction enabled covalent conjugation of the C_{60} moiety with the pendant aldehyde group containing copolymer **6**. Taken together, the evidence from the FTIR, ^1H -NMR, and ^{13}C -NMR spectroscopies corroborated the successful synthesis of polythiophene/ C_{60} D–A double cable polymer.

The UV/Vis absorption spectra of copolymer **6**, covalent hybrid **7**, D–A physical hybrid **8** and pristine C_{60} recorded using 0.5 mg mL^{-1} CHCl_3 solutions are depicted in Fig. 3a. The UV/Vis absorption spectrum of C_{60} showed a narrow absorption peak with λ_{max} at 329 nm. Polymer **6** exhibited a UV/Vis profile containing an absorption peak at λ_{max} 430 nm, which is a characteristic of polythiophenes. The UV/Vis spectrum of covalent hybrid **7** showed two distinct absorption signals. The absorption at λ_{max} 440 nm corresponds to the absorbance due the polythiophene backbone while λ_{max} 329 nm originates from the absorbance due to the

covalently conjugated fullerenes in covalent hybrid **7**. As in the case of covalent hybrid **7**, physical hybrid **8** also showed two significant absorption signals corresponding to physically blended P3HT and C_{60} (Fig. 3a). The solutions used for recording UV/Visible absorption spectra were further characterized by fluorescence spectroscopy. The fluorescence spectra of polymer **6**, covalent hybrid **7**, and physical hybrid **8** were recorded after excitation at 450 nm. Fluorescence quenching can be used to gauge the extent of interaction between electron donor and acceptor species.^{52,54} For the solutions containing equivalent molar concentrations of donor and acceptor, the covalent D–A hybrid **7** exhibited a higher fluorescence quenching than the D–A physical hybrid **8** when compared to the fluorescence of the random copolymer **6** (Fig. 3c). Such an observation can be attributed to the close molecular level D–A proximity in the case of covalent hybrid **7**, that facilitated the electron transfer between donor and acceptor molecular components. The donor molecular component of covalent hybrid **7** absorbs energy upon excitation that results in the excitation of electrons from HOMO to the LUMO of a donor molecular component. Compared to the physical D–A hybrid **8**, the excited electrons upon relaxation are more efficiently accepted into the LUMO of the molecular acceptor component of the covalent hybrid **7**. This resulted in a higher extent of fluorescence quenching observed for the covalent D–A hybrid **7**. The UV/Vis absorbance and fluorescence spectra were also recorded after diluting the original 0.5 mg mL^{-1}

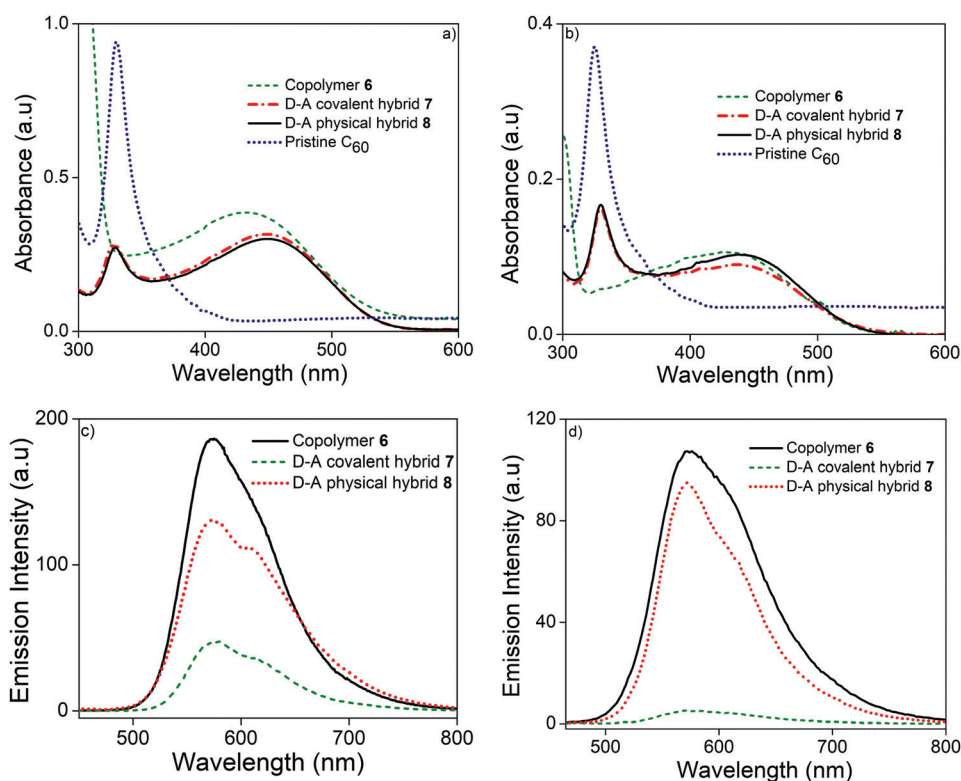


Fig. 3 UV/Vis absorption spectra of copolymer **6**, D–A covalent hybrid **7**, D–A physical hybrid **8** and pristine C_{60} (a) recorded on solutions of 0.5 mg mL^{-1} concentration, (b) recorded on solutions after 100-fold dilution; fluorescence emission spectra of random copolymer **6**, D–A covalent hybrid **7** and D–A physical hybrid **8** (excitation wavelength: 450 nm) (c) recorded on solutions of 0.5 mg mL^{-1} concentration and (d) recorded on solutions after 100-fold dilution.

CHCl_3 solutions and the obtained spectra are shown in Fig. 3b and d. The characteristic absorbance signals for copolymer and C_{60} can be clearly seen in the UV/Vis absorbance spectra of the diluted solutions of copolymer 6, covalent hybrid 7, physical hybrid 8 and pristine C_{60} (Fig. 3b). In addition, the higher extent of the fluorescence quenching in the case of the covalent hybrid 7 is evident from the fluorescence emission spectra of the diluted solutions depicted in Fig. 3d. This suggests that the higher extent of fluorescence quenching observed in the case of covalent hybrid 7 is indeed due to the close molecular level D–A proximity that facilitated the electron transfer between donor and acceptor molecular components.

In order to further demonstrate the efficient D–A electron transfer in the case of covalent hybrid 7, three separate photodetectors were fabricated using random copolymer 6, covalent hybrid 7 and physical hybrid 8, and their current–voltage (I – V) curves were recorded in the dark and under exposure to white light (Fig. 4). All measurements were performed at room temperature. The photoactive layers (thickness 200–300 nm) of random copolymer 6, covalent hybrid 7, and physical hybrid 8 were spin coated (2000 rpm) on the interdigitated Au electrodes to fabricate functional photodetectors. Upon recording the I – V profiles of polymer 6, covalent hybrid 7, and physical hybrid 8 under dark and light conditions between -5 V and $+5$ V, it was

observed that these materials have an extremely small dark current of about $0.003 \mu\text{A}$ at 5 V bias. Upon exposure to white light, the current was observed to increase to $0.05 \mu\text{A}$ and $0.07 \mu\text{A}$ at 5 V bias in the case of the photodetector fabricated with polymer 6 and its physical hybrid 8. On the other hand, the photodetector fabricated with the covalent hybrid 7 as a photoactive layer exhibited a much enhanced photocurrent of $0.18 \mu\text{A}$ at 5 V bias. Compared with the devices fabricated with physical hybrid 8 and donor polymer 6 as photoactive layers, the device fabricated with the covalent hybrid 7 showed about 2.5 times and 3.6 times enhancement, respectively, in photocurrent generation at the equivalent bias. We attribute the generation of significantly higher photocurrent in the case of the device containing covalent hybrid 7 in the active layer to the covalent association between donor and acceptor molecular components. This proof of concept photoelectrical characterization validates our hypothesis by revealing the superior photocurrent generation capability of the D–A covalent hybrid. In order to further improve the performance, future work will focus on the optimization of device architecture, film thickness, and application of annealing. In addition detailed evaluation of electrical and spectroscopic parameters is also warranted.

Moreover, it can be observed (Fig. 4b) that the current is not zero at zero voltage. This is because of the fact that the photodetectors have a reversed biased junction when connected to a battery. The polarity of the connections causes electrons and holes to move away from the junction, which reinforces the potential barrier and increases the depletion region. Ideally, when the voltage is zero, the current should also be zero. However, a small current at zero voltage was observed in some of the devices due to a small number of electron–hole pairs generated by thermal agitation of atoms of the material that may flow even when the voltage is zero.^{62–64} From I – V curves, it can be inferred that the electron–hole pairs (excitons) generated upon exposure to light were most effectively dissociated with lesser probability of recombination in the case of the covalent photodetector fabricated with the active layer constituted by D–A covalent hybrid 7. These observations reflect that the dissociation of excitons into electrons and holes and their mobility towards the respective electrodes to generate photocurrent is most likely facilitated by the close proximity of C_{60} , which has high affinity towards electrons, covalently held close to the donor polythiophene component of D–A covalent hybrid 7.

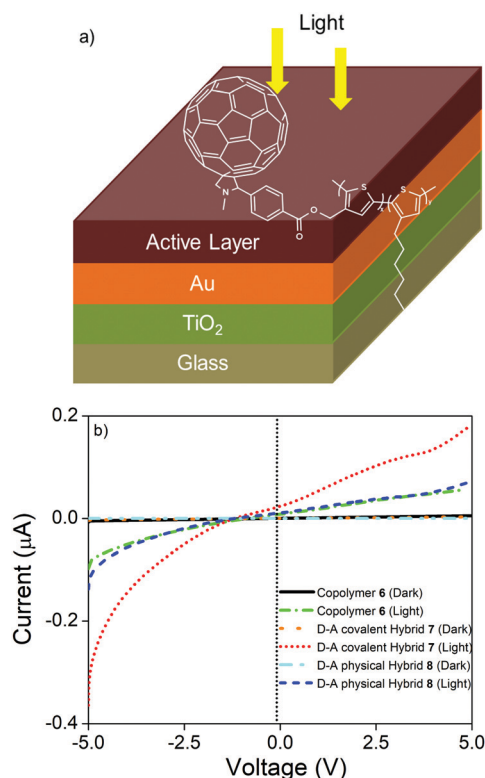


Fig. 4 (a) Schematic representation of the photodetector device structure fabricated with the active layer constituted by the D–A covalent hybrid 7. (b) Comparison of the I – V profiles (in dark and upon illumination with white light at room temperature) recorded for the photodetectors comprising the active layers constituted by the copolymer 6, D–A covalent hybrid 7 and D–A physical hybrid 8.

Conclusions

A polythiophene copolymer with pendant aldehyde groups was synthesized using GRIM polymerization followed by side chain engineering employing controlled chemical modifications. The synthesized side chain engineering electron donor OSP was conjugated to the pristine C_{60} electron acceptor *via* the Prato reaction. The resulting covalent D–A hybrid exhibited higher fluorescence quenching as a result of the close D–A proximity and efficient D–A electronic communication when compared to the physical D–A hybrid. This efficient electronic communication

further translated into higher photocurrent generation in the photodetector devices fabricated with the covalent hybrid as the photoactive layer. The results reported in this study constitute a step in the systematic development of D–A covalent hybrids with controlled molecular design to fine tune and control the photo-voltaic effect.

Conflicts of interest

There are no conflicts to declare.

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